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D.J. Robertson

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A paleomagnetic study of Rangitoto Island, Auckland, New Zealand

D. J. ROBERTSON
Department of Physics
University of Auckland
Private Bag
Auckland, New Zealand*

Abstract A paleomagnetic study has been made of 99 specimens from 11 sites in the recent basalt lavas of Rangitoto Island. The measured natural remanent magnetisation (NRM) had a formation mean direction of 1.9°E, −62.1°, α_{95} = 2.6°; thermal cleaning changed the formation mean NRM direction only slightly to 0.2°E, −61.7°, α_{95}= 1.8°. The densities, magnetisations, and magnetic susceptibilities measured for each site are recorded. Since Königsberger's Q_n ratio is large in every case, it follows that the aeromagnetic total force anomaly is symmetrical about the geographic meridian. Paleomagnetic declinations, at each site, when compared with historically recorded declinations of the geomagnetic field, confirm the dating results previously obtained by the ^{14}C method; that is, Rangitoto was an active volcano between A.D. 850 and A.D. 1800, with a maximum of activity between A.D. 1200 and A.D. 1500. It is inferred that near Auckland the geomagnetic field has changed direction at approximately 2.87×10^{-2} deg/year eastwards, and changed little in inclination, during the last 1000 years.

Keywords paleomagnetism; magnetic declination; age determination; Rangitoto Island

INTRODUCTION

Rangitoto Island (36°47′S, 174°52′E) is a symmetrical volcanic cone (Fig. 1), 6 km in diameter and 260 m high. Its structure has been elucidated by the geophysical study of Milligan (1977), and its geology has been summarised by Ballance & Smith (1982). The summit is formed by a cluster of scoria cones, the highest of which has a crater 300 m in diameter and 75 m deep. The boundary of the scoria cones is outlined by a small depression, or "moat structure", which approximately follows the 140 m contour. Below the 140 m contour, a glacis of vesicular pahoehoe and aa lava flows follows a gentle slope to the sea. Geophysical studies (Milligan 1977) indicate that the basalt is underlain by low-density scoria.

Lava flows were sampled for paleomagnetic study at 11 sites, which are shown in Fig. 1. Sites 1101, 1102, 1103, 1104, 1108, and 1111 are in lava flows exposed in old quarries. At sites 1105, 1107, 1109, and 1110, samples were taken from the selvedge of pahoehoe lava flows, which at site 1109 forms a baked contact with fossiliferous mudstone. At site 1106, the highest, and probably the youngest, basalt flow was sampled where it rests on scoria near the edge of the central scoria cone.

Petrographic descriptions prepared by P. M. Black, pertaining to each site, are set out in Robertson (1983). The rock is mainly a crystalline olivine basalt, with phenocrysts of olivine and microphenocrysts of augite and plagioclase in a groundmass of augite, feldspar, glass, and opaque minerals smaller than 50 μm in diameter. Specimens from sites 1107, 1109, and 1110 are particularly glassy, which is consistent with their proximity to chilled margins, and the specimen from site 1109 contained no opaque minerals large enough to be observed with the optical microscope.

For the paleomagnetic analysis, true north was established at each site by resection, or else by the method of hour angle of the sun. Hence, the geomagnetic declination at each site could be determined with a prismatic compass, and this parameter is listed in Table 1.

PALEOMAGNETIC ANALYSIS

For the 11 sites shown in Fig. 1, 99 oriented specimens were analysed. Their physical characteristics are listed in Appendix 1.

In particular, the observed mean particle density (2.91 ± 0.06 Mg/m^3) is typical of fresh basalt, and the average void ratio (13.2 ± 3.2%) is high, due to the notable vesicularity. The magnetic susceptibility ($6.5 \pm 4.8 \times 10^{-3}$ SI) is not exceptional, but the

*Present address: Department of Earth Sciences, University of Leeds, Leeds LS2 9JT, United Kingdom.
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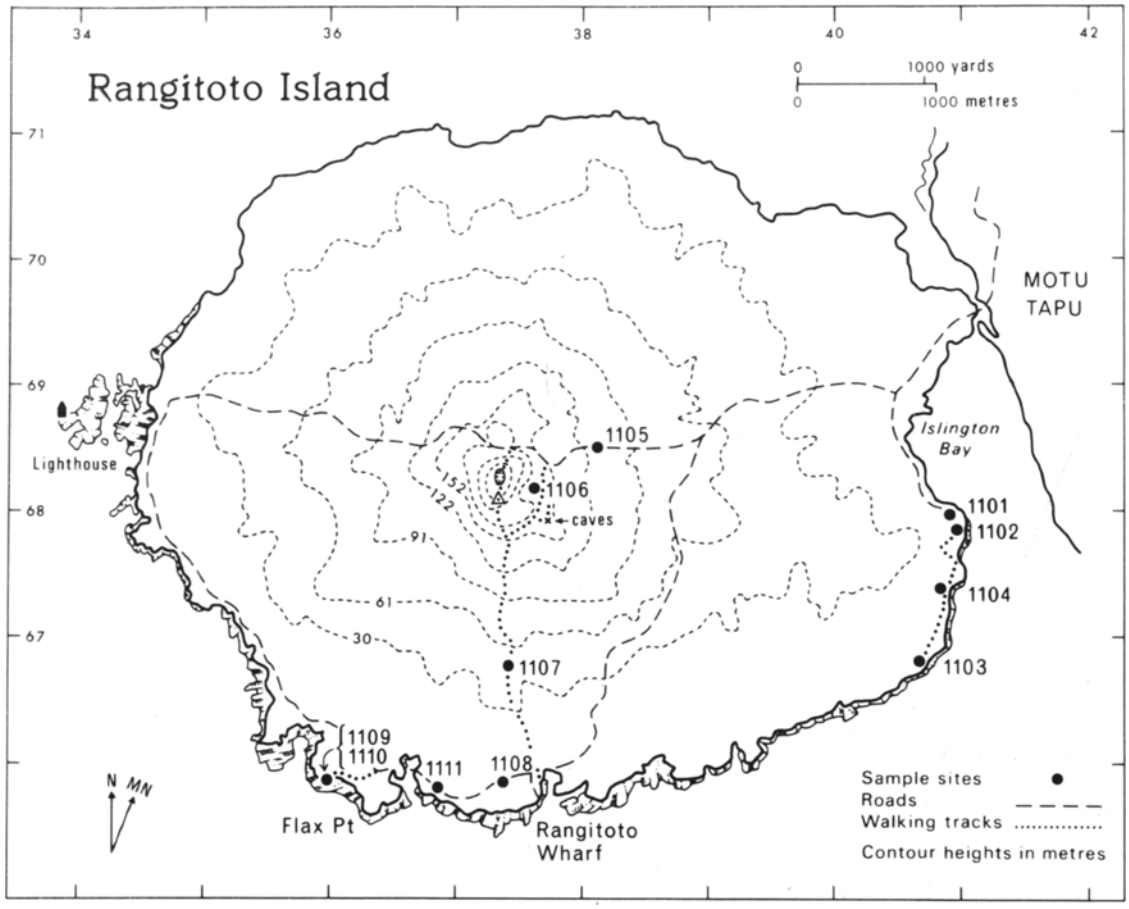


Fig. 1 Rangitoto Island, Waitemata Harbour, Auckland, showing location of sample sites used in this study. (Graticule: New Zealand national yard grid, NZMS 1 sheet N42, Auckland).

mean magnetisation (11.9 ± 8.0 A/m) is unusually high. Consequently, the Q_n ratio (46 ± 24) (Königsberger 1938) is also very high. The total magnetisation of the rock *in situ* is therefore dominated by remanent, rather than by induced, magnetisation.

For the two sites at Flax Point, 1109 and 1110, the Q_n ratio is extremely high (100 ± 19, and 83 ± 15 respectively). This is consistent with small magnetite grains being formed, such as by quenching of the lava in water (cf. Hatherton 1954), which is entirely consistent with the site and petrographic descriptions above. Because these are the only intertidal sites included in the study, it follows that the sea level is approximately the same now as it was when these lava flows solidified. Site mean natural remanent magnetisation (NRM) directions, obtained by applying the statistical methods of Fisher (1953), are shown in Table 1. They are closely grouped about the geographic meridian. If the site mean directions are themselves averaged using Fisher's statistics (a "two-tier analysis"), then the formation mean NRM direction is found to be 1.9°E, −62.1° (i.e. 62.1 upwards, using the convention established in the Northern Hemisphere, whereby upward magnetic inclination has negative sign), and the semivertical angle of the cone of 95% confidence in the mean direction (α_{95}) is 2.6°. Since $Q_n \gg 1$, this is consistent with the total force aeromagnetic anomaly being symmetrical about the true, rather than magnetic, north direction, as was observed by Milligan (1977).

One pilot specimen from each site was subjected to progressive thermal demagnetisation, with steps of 100°C. The resulting demagnetisation trajectories were all linear, and an example is shown in the form of a modified Zijderveld diagram in Fig. 2.

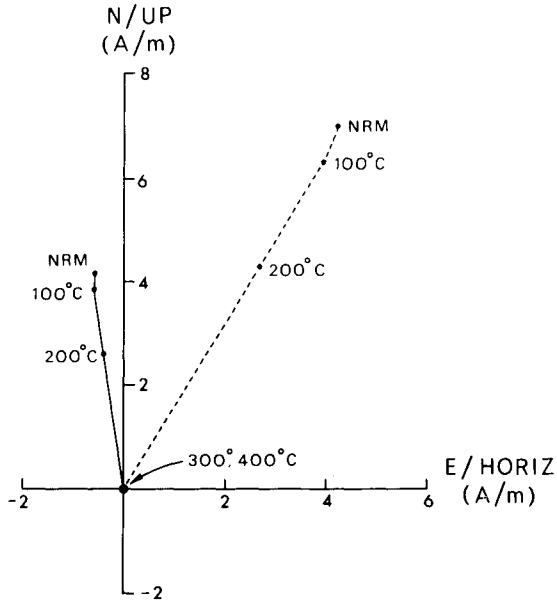


Fig. 2 Progressive thermal demagnetisation trajectory for specimen 1542.1, from site 1104, shown on a modified Zijderveld diagram. The solid line is the locus of the horizontal projection of the remanent magnetisation vector after each heating cycle, and gives declination directly. The dotted line, which relates the vertical to horizontal component, gives inclination directly.

The intensity of magnetisation falls quickly, as shown on Fig. 3, and most of the specimens are almost completely demagnetised at 300°C.

On the basis of the pilot demagnetisation results, approximately six specimens per site were thermally cleaned at a temperature selected to remove about 60% of the NRM. The resulting site mean thermally cleaned NRM directions are summarised in Table 1, and shown in Fig. 4. The dispersions of the cleaned NRM directions are comparable with those of the original NRM, and very few site mean directions have changed significantly. The formation mean of the vector difference between the original NRM and the thermally cleaned NRM for each site is 1.5°E, −62.5°, α_{95} = 7.1°. This is very close to both the original and cleaned NRM directions, and therefore implies that the NRM is stable in direction and has simply been decreased in magnitude.

The formation mean cleaned paleofield direction is 0.2°E, −61.7°, α_{95} = 1.8°. This is similar in inclination, but significantly to the west of the geomagnetic field direction, which was 19.0°E, −62.7° at the time of sampling (L. A. Tomlinson, Geophysical Observatory, DSIR, pers. comm.). This result is not surprising, given that Rangitoto is of recent genesis, and that the geomagnetic field in New Zealand is increasing in easterly declination and (much less rapidly) in upward inclination, at the present time (L. A. Tomlinson, pers. comm.).

Table 1 Summary of paleofield directions.

			NRM				Thermally cleaned			
Site no.	GR	CV (°)	N	Dec (°)	Inc (°)	α_{95} (°)	T (°C)	N	Dec (°)	Inc (°)	α_{95} (°)
1101	409680	19.6	4	359.1	−66.2	4.6	200	4	358.7	−62.9	5.4
1102	410679	17.3	9	001.7	−58.3	1.4	200	6	004.3	−61.1	3.4
1103	408668	17.2	10	002.7	−62.3	1.4	175	6	357.9	−61.6	2.5
1104	409675	19.0	10	353.9	−62.0	3.2	200	6	349.9	−64.8	4.9
1105	381685	24.2	11	005.2	−70.5	3.1	180	6	358.8	−61.9	3.6
1106	376682	21.4	9	012.7	−61.8	2.7	200	5	013.2	−63.1	4.2
1107	374668	20.6	7	356.7	−56.0	13.6	200	5	354.3	−61.5	10.2
1108	374659	21.7	5	360.0	−63.4	2.3	140	4	001.6	−62.2	1.0
1109	359659	20.2	8	004.0	−58.6	1.9	100	5	002.5	−58.8	2.7
1110	359659	21.1	8	001.1	−62.0	2.7	75	6	357.9	−60.9	3.0
1111	369657	22.8	11	005.1	−61.1	1.8	200	4	002.7	−59.2	4.0
Formation mean		20.5		001.9	−62.1	2.6			000.2	−61.7	1.8

GR = Grid references based on the thousand-yard grid of the topographic map series (NZMS 1) 1:63 360, sheet N42— Auckland.

CV = Compass variation (i.e. measured geomagnetic declination (east) at approximately 1.5 m above ground level at each site).

T = Temperature chosen for thermal cleaning.

Formation mean = directions obtained by a two-tier Fisher averaging procedure, in which all sites have been accorded equal weight.

Upwards directed inclination has negative sign.

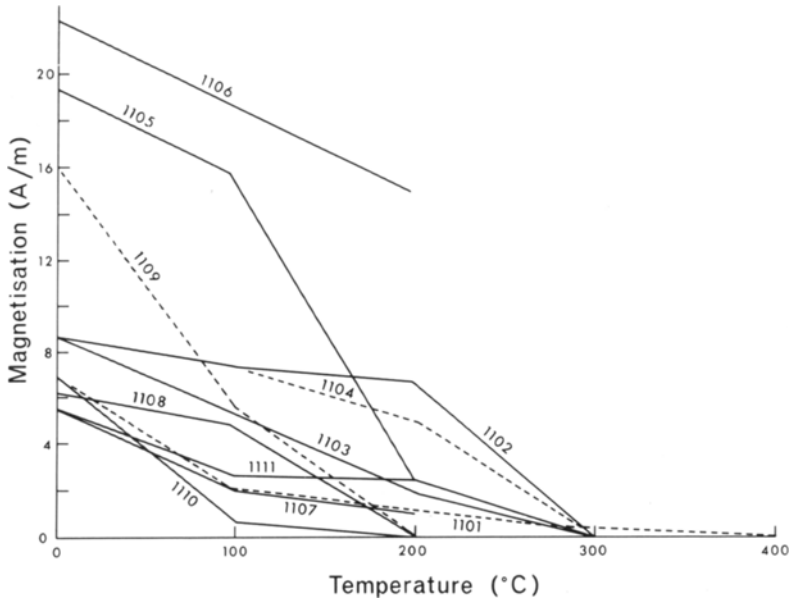


Fig. 3 Thermomagnetic curves for one pilot specimen from each site.

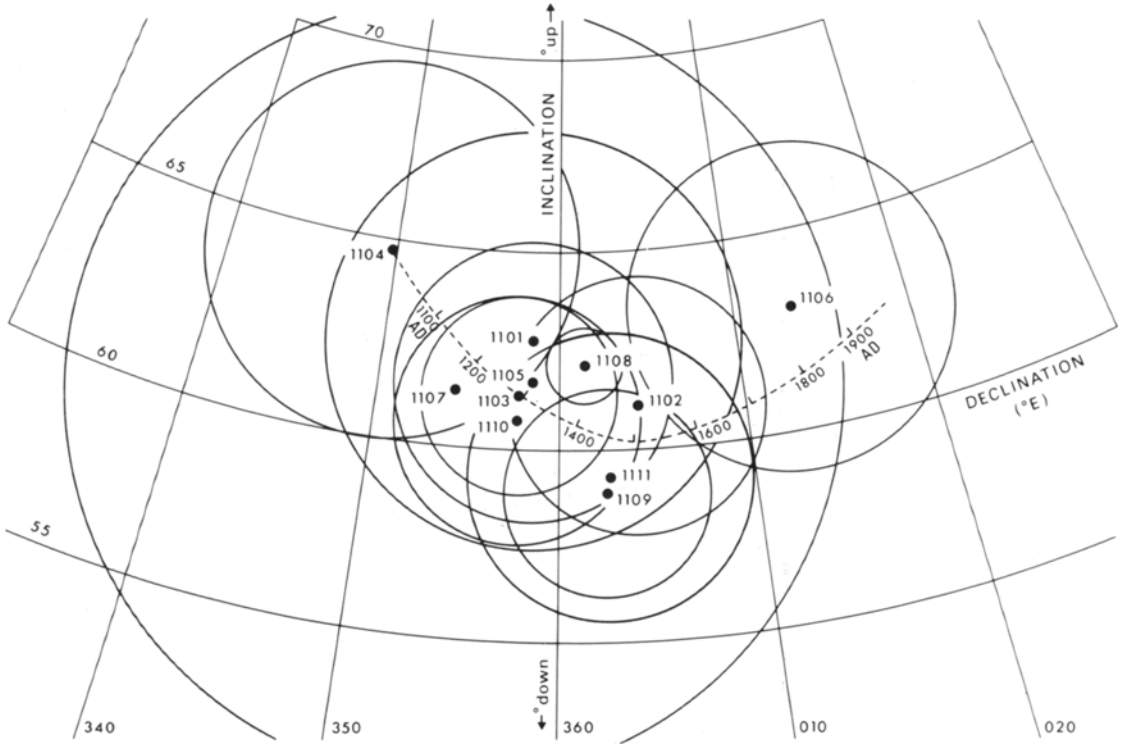


Fig. 4 Site mean directions for NRM after thermal cleaning has been carried out. The circles represent α_{95}. (Graticule: Orthomorphic conic projection, standard at 60°).

CHRONOLOGY

There is no historical record of eruptions of Rangitoto, although culture layers of human habitation are overlain by ash on the adjoining island of Motutapu. Millener (1979) stated that the oldest trees on Rangitoto are < 200 years old. This opinion is supported by referring to a photograph by Dr Kinder, dated 1869, which shows virtually no vegetation on the scoria cone, and quoting Sarah Mathew, who, in her journal, described walking through fine ash to the summit in 1841.

Twelve radiometric (^{14}C) age determinations pertaining to eruptions of Rangitoto have been summarised by Law (1975) (see Fig. 6). The ^{14}C dates for marine carbonate materials provide very conservative lower bounds on the events concerned, because the marine carbon may have been out of the atmospheric carbon cycle for some time before it became incorporated in the shells, and moreover the shells may have been quite old before becoming overlain by the ash or lava. Nevertheless, the eruptive history of Rangitoto Island is bracketed by ^{14}C dating to the interval from 1100 B.P. to 190 B.P. (i.e. A.D. 850 to A.D. 1760).

The radiometric geochronology of Rangitoto can now be compared with the magnetic record. A link between the two is indicated by historic records of the geomagnetic declination.

The geomagnetic declination in the vicinity of the northern North Island was determined with great care by each of the explorers Tasman (1642) and Cook (1770). To their data can be added the more recent figures of Felton-Mathew (1841), and the present-day field quoted above, to produce the graph, Fig. 5. The regression of declination against time is remarkably linear and is given by:

$$D(Y) = 2.87 \times 10^{-2}Y - 38.70, \quad (1)$$

where D is the declination in degrees, and Y is the year A.D. Hence:

$$Y(D) = 34.88 \times D + 1351. \quad (2)$$

Equation 2 can be extrapolated only with caution. The rate of change in D is unlikely to always have been so constant, and, at this latitude, D seldom lies outside of the range from 340° to 020° E. Nevertheless, a current study by the author indictes that the trend of equation 2 has prevailed at least since the rhyolites of Mount Tarawera were formed during the Kaharoa event approximately 700 years ago.

Utilising equation 2, an approximate relationship between paleofield declination, inclination, and time has been sketched through the data points of Fig. 4. This relationship is potentially useful for archeomagnetic dating and for testing models of the geomagnetic field.

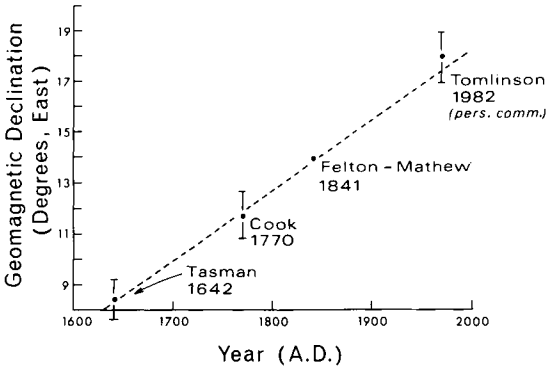


Fig. 5 Geomagnetic declination for Waitemata Harbour, extracted from historical records. For Tasman's and Cook's data an average value at Rangitoto is estimated by averaging their various observations between 34°S and 36°S latitude. The error bars show the range of measurements recorded over this region.

Assuming that the change in D was linear prior to A.D. 1642, an approximate archeomagnetic age for each site can be deduced from the site mean declination listed in Table 1, with an error estimated from the α_{95} limits. Archeomagnetic age estimates obtained in this way for the lava flows which were sampled are shown in Fig. 6, in comparison with the published ^{14}C dates. The archeomagnetic date for the site 1106 is unreliable, because the outcrop may have subsided, although it is the youngest flow stratigraphically. The archeomagnetic date for site 1104 is consistent with its stratigraphically old position at the base of a large quarry.

There is general agreement between the spread of dates obtained archeomagnetically for Rangitoto Island lavas and those obtained by ^{14}C dating of material associated with ash showers. A cluster of archeomagnetic date estimates around A.D. 1300 may correspond to a similar cluster of ^{14}C estimates at about A.D. 1320. Thus, Rangitoto appears to have been an active volcano between A.D. 850 and A.D. 1800 with a maximum of lava flow activity around A.D. 1300.

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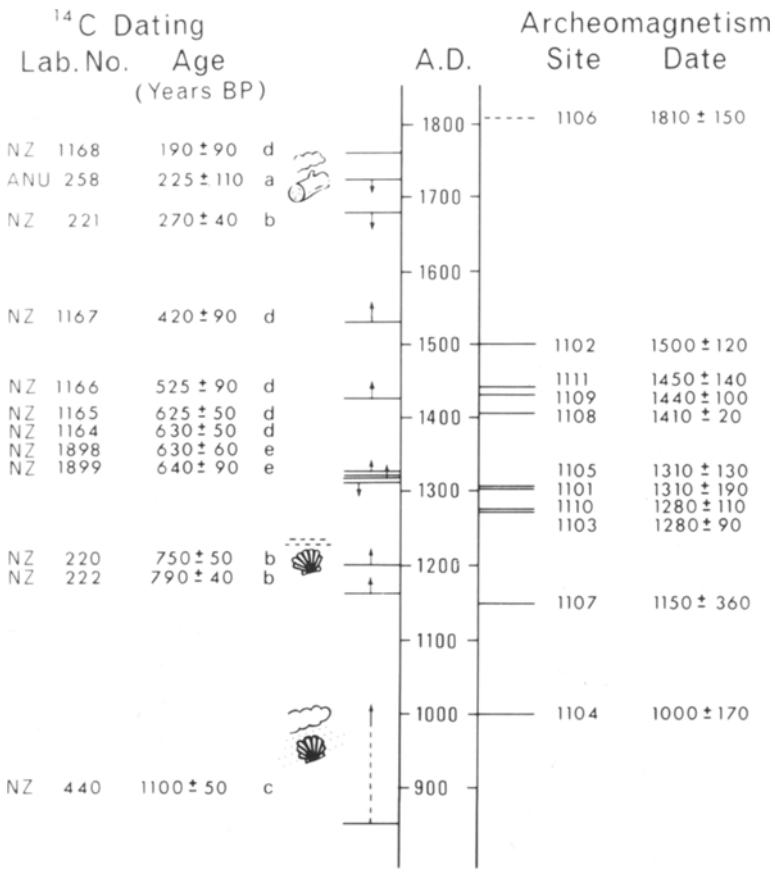


Fig. 6 Chronology of Rangitoto Island. ^{14}C data are taken from Law (1975), a compilation from: (a) Polach et al. (1969), (b) Brothers & Golson (1959), (c) Grant-Taylor & Rafter (1971), (d) Davidson (1972), and (e) Green, unpublished data. The ^{14}C ages for the "NZ" numbered specimens differ slightly (by at most 50 years) from those quoted by Law (1975), since they have recently been recomputed (M. R. Manning, Institute of Nuclear Sciences, DSIR, pers. comm.) using revised standards and a 5730 year half-life for ^{14}C, to obtain a best estimate of the true age in calendar years. Most of these determinations relate to burnt wood associated with volcanic ash on Motutapu and provide upper or lower bound on the age of ash showers as indicated by small arrows. Dates without arrows apply to charcoal from cooking pits. The pictorial symbols indicate wood found beneath a lava flow on Rangitoto, sea shells underlying ash on Motutapu, and sea shells in baked contact with lava at Flax point.

REFERENCES

Ballance, P. F.; Smith, E. M. 1982: Walks through Auckland's geological past. *Geological Society of New Zealand guide book 5.*

Brothers, R. N.; Golson, J. 1959: Geological and archeological interpretation of a section in Rangitoto Ash on Motutapu Island. *New Zealand journal of geology and geophysics 2*: 569–577.

Cook, J. 1770: Chart of New Zealand. *In:* Beaglehole; J. C. The journals of Captain James Cook on his voyages of discovery; XII. Cambridge University Press, 1955.

Davidson, J. M. 1972: Archeological investigations on Motutapu Island, New Zealand. *Records of the Auckland Institute Museum 9*: 1–14.

Felton-Mathew 1841: Map of the harbour of Waitemata. New Zealand, Department of Lands and Survey.

Fisher, R. A. 1953: Dispersion on a sphere. *Proceedings of the Royal Society, London A217*: 295–303.

Grant-Taylor, T. L.; Rafter, T. A. 1971: New Zealand radiocarbon age measurements—6. *New Zealand journal of geology and geophysics 14*: 364–402.

Hatherton, T. 1954: The permanent magnetisation of horizontal volcanic sheets. *Journal of geophysical research 59*: 223–232.

Königsberger, J. G. 1938: Natural residual magnetism of eruptive rocks, parts I and II. *Terrestrial magnetism and atmospheric electricity 43*: 119–127, 229–320.

Law, R. G. 1975: Radiocarbon dates for Rangitoto and Motutapu, a consideration of dating accuracy. *New Zealand journal of science 18*: 441–451.

Millener, L. H. 1979: Forest, scrub and fresh water communities. *In:* Brook, P. J. *ed.* Natural history of Auckland; an introduction. Auckland War Memorial Museum handbook. Pelorous Press, Auckland, New Zealand.

Milligan, J. 1977: A geophysical study of Rangitoto volcano. Unpublished M.Sc. thesis, lodged in the Library, University of Auckland.

Polach, H. A.; Chappell, J.; Lovering, J. F. 1969: ANU radiocarbon date list III. *Radiocarbon 11(2)*: 245–262.

Robertson, D. J. 1983: Paleomagnetism and geochronology of volcanics in the northern North Island, New Zealand. Unpublished Ph.D. thesis, lodged in the Library, University of Auckland.

Tasman, A. J. 1642: Journal. Published in facsimile, with translation by J. De Hoop Scheffer; and magnetic observations by W. Van Bemmelen, Amsterdam, F. Muller and Co. 1898; reprinted Los Angeles, N. A. Kovach, 1965.

Appendix 1 Physical characteristics of the recent basalt lavas sampled in this study.

Site no.	No. of specimens	ρ_{dry} (Mg/m^3)	ρ_{wet} (Mg/m^3)	$\rho_{particle}$ (Mg/m^3)	Void ratio (%)	χ (SI$\times 10^{-2}$)	M (A/m)	Q_n
1101	5	2.64	2.75	2.95	10.48	0.404	5.26	29.5
		±0.08	±0.06	±0.02	±2.58	±0.029	±1.69	±9.5
1102	9	2.73	2.81	2.96	7.89	0.478	4.81	22.2
		±0.11	±0.09	±0.04	±3.22	±0.045	±2.29	±8.9
1103	12	2.72	2.82	3.01	9.85	0.514	9.17	37.7
		±0.08	±0.06	±0.02	±2.74	±0.142	±5.70	±14.4
1104	10	2.57	2.70	2.96	13.31	0.667	18.47*	52.1
		±0.11	±0.09	±0.05	±2.53	±0.348	±16.91	±25.3
1105	11	2.50	2.64	2.92	14.4	1.661	24.54	34.4
		±0.05	±0.03	±0.04	±2.23	±1.029	±15.27	±7.6
1106	9	2.39	2.58	2.94	18.54	1.523	26.7	43.4
		±0.07	±0.05	±0.02	±2.10	±0.636	±7.7	±12.9
1107	8	2.42	2.57	2.83	14.41	0.451	6.59	34.6
		±0.05	±0.04	±0.02	±1.59	±0.088	±1.45	±11.9
1108	5	2.47	2.62	2.89	14.34	0.364	5.22	32.4
		±0.02	±0.02	±0.00	±0.76	±0.023	±0.69	±4.9
1109	8	2.58	2.69	2.88	10.36	0.330	14.31	99.9
		±0.10	±0.08	±0.05	±2.47	±0.045	±1.44	±19.0
1110	8	2.43	2.57	2.83	14.09	0.204	7.26	83.2
		±0.10	±0.07	±0.03	±2.56	±0.056	±0.82	±15.0
1111	14	2.34	2.51	2.83	17.20	0.534	8.53	38.2
		±0.08	±0.06	±0.03	±1.90	±0.144	±3.79	±23.3
Mean	11 sites	2.53	2.66	2.91	13.17	0.648	11.89	46.5
	99 specs	±0.13	±0.10	±0.06	±3.23	±0.483	±7.96	24.1

*Bimodal.
± figures are the within-site standard deviation.
χ is the volume susceptibility, which is dimensionless in the SI system of units.
M = natural remanent magnetisation.
$Q_n = (\mu_o M)/(\chi B_o)$, where $\mu_o = 4\pi \times 10^{-7}$ and $B_o = 56\,000$ nT.